

A Lean Transcription of the Corrected Lazard Formula for Solvable Quintics

ProveIt project

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Abstract

Daniel Lazard gave an explicit radical formula for every irreducible solvable quintic. This note documents a Lean 4 transcription of the formula, including Vladimir Reshetnikov's correction of two terms in the printed expression P_{22} and Valeriy Zapodovnikov's restoration of the exceptional sign choice needed when the first determination of ε would make $T = 0$.

The Lean development formalizes the conditional data flow after the required algebraic witnesses have been supplied: normalization and depression, Lazard's degree-six resolvent, the four equations recovering $i_4, \dots, i_8$, all auxiliary invariants $D, \dots, K$, both coherent sign-selection stages, the corrected P_{22} , and inverse Fourier reconstruction of five candidate outputs. It proves the normalization identity, the equivalence of the monic and fraction-free resolvent equations, preservation of ε^2 by the correction, nonvanishing consequences of the branch certificates, coherence of all four quadratic sign branches, and soundness of finite branch selection. It does *not* prove that the five displayed outputs are roots, nor that the needed certificates exist. Pointwise soundness and full five-factor completeness are recorded as separate propositions, not as axioms.

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1 Purpose, sources, and claim boundary

The formal source is

`Algebra/PolynomialFormulas/Lean/PolynomialFormulas/LazardQuintic.lean`.

It is imported by the public `PolynomialFormulas` module. The source materials were reviewed at Smithereens commit `da7026ec1369ade49bb6d964019a4c124fdd5091`. The principal references are:

- (1) Lazard’s chapter *Solving Quintics by Radicals*, especially the explicit formula on printed pp. 219–222 [1];
- (2) Lazard’s original Maple worksheet, version 3 (March 2002), distributed with those materials;
- (3) Reshetnikov’s Wolfram-language transcription and warning about the two printed P_{22} errors [4];
- (4) Zapodovnikov’s corrected 2025 version on the same page, especially `If[(g+h/F)==0,F=-F]`; and
- (5) Dummit’s independent solvability resolvent [2], together with Chapman’s repair of the specialized-distinctness step [3].

The expanded review also covered every text-readable document under `src/Polynomial` whose path or contents mention “quintic”, including the Reshetnikov notebooks, the Wolfram notebooks, the independent `solving-quintics` package, Dummit’s paper, and the corpus reports. These additional files provide useful examples and independent checks, but they do not supersede Lazard’s formula. In particular, the `solving-quintics` package switches to numerical root matching after the resolvent and is not an exact implementation of the formula documented here.

The present claim is deliberately limited:

The Lean definitions are an exact, type-correct transcription of the corrected algebraic algorithm, organized so that its eventual correctness proof can be stated without changing the API. No theorem in this contribution asserts the unproved final root identity.

This distinction matters. Lean accepts division by zero as a total field operation, and so a raw expression alone would not faithfully express Lazard’s partial algebraic construction. The formalization therefore records every required denominator condition and every shared radical determination in `RadicalCertificate`.

2 Mathematical scope

Lazard begins with an irreducible quintic over a field of characteristic different from 2 and 5. His compact expanded formula uses a particular fourth projection whose invertibility is justified when -1 is not a square; this is harmless over $\mathbb{Q}$. The paper also sketches an alternate projection for more general fields, but does not print its expanded formula.

The Lean expressions are polymorphic over a field K with `CharZeroK`. This is a slightly stronger characteristic assumption and ensures that all displayed rational denominators are meaningful. The actual rational front end is made explicit by `RationalInvariantCertificate`: the tuple $(i_4, \dots, i_8)$ is first certified in $\mathbb{Q}$ and only then mapped to a radical extension. The abstract generic API remains useful for algebraic lemmas.

The generic propositions `FormulaSound` and `FormulaFactors` are deliberately stronger *conditional algebraic targets*: they quantify over an arbitrary characteristic-zero field, but only after every displayed invariant and radical certificate has been supplied. They do not assert existence of those certificates. Thus Lazard’s sourced irreducibility and -1 nonsquare hypotheses remain obligations for the applicability/existence bridge, while the rational specialization fixes the coefficient field to $\mathbb{Q}$ and performs radical calculations in an extension field.

For a complete solver on arbitrary quintics, reducible inputs must be factored first. Every proper factor has degree at most four and can in principle be handled by the existing low-degree formulas. This module implements neither that factor search nor the rational-root search for the sextic: its rational entry point accepts a supplied `RationalInvariantCertificate`. Lazard’s core is the irreducible branch; the formalization does not misrepresent it as a total reducer for arbitrary degree-five polynomials.

3 The algorithm at a glance

For an irreducible rational quintic, the corrected construction is:

1. normalize and translate to $Y^5 + pY^3 + qY^2 + rY + s$;
2. find the rational root i_4 of Lazard’s sextic resolvent;
3. solve the four linear equations of Figure 3 for i_5, i_6, i_7, i_8 ;
4. calculate D, E, F, G and choose $\varepsilon^2 = 5D$;
5. if $E + F/\varepsilon = 0$, replace ε by $-\varepsilon$;
6. choose $T^2 = \frac{5}{2}(E + F/\varepsilon)$ and set $U = 5G/(T\varepsilon)$;
7. among four coherent sign triples (ε, T, U) , choose one for which $Q_1 \neq 0$;
8. choose one shared fifth root $P_1^5 = Q_1$, calculate the corrected P_2, P_3, P_4 , and apply the inverse discrete Fourier transform;
9. translate every resulting root back by $-b/(5a)$.

The group-theoretic tower behind these steps is

$$C_5 \subset D_5 \subset F_{20} \subset S_5, \quad [D_5 : C_5] = 2, \quad [F_{20} : D_5] = 2, \quad [S_5 : F_{20}] = 6.$$

The sextic selects a conjugate of the maximal solvable subgroup F_{20} ; the two square roots descend through the two index-two inclusions; one fifth root resolves the remaining cyclic action.

4 Normalization and depression

Write

$$f_0(X) = aX^5 + bX^4 + cX^3 + dX^2 + eX + f, \quad a \neq 0.$$

Under $X = Y - b/(5a)$ and division by a , one obtains

$$f(Y) = Y^5 + pY^3 + qY^2 + rY + s,$$

where

$$\begin{aligned} p &= \frac{5ac - 2b^2}{5a^2}, \\ q &= \frac{25a^2d - 15abc + 4b^3}{25a^3}, \\ r &= \frac{125a^3e - 50a^2bd + 15ab^2c - 3b^4}{125a^4}, \\ s &= \frac{3125a^4f - 625a^3be + 125a^2b^2d - 25ab^3c + 4b^5}{3125a^5}. \end{aligned}$$

These are **depress**. Lean proves, for every Y ,

$$f_0\left(Y - \frac{b}{5a}\right) = a f(Y)$$

as **depress_eval**. It also proves that depression commutes with a field homomorphism (**depress_map**), which is the bridge from rational certificates to the radical field.

5 Fourier invariants and metacyclic data

Let $x_0, \dots, x_4$ be an ordering of the roots and let ω be a primitive fifth root of unity. Lazard uses the Fourier sums

$$s_k = \sum_{j=0}^4 \omega^{kj} x_j, \quad k = 0, \dots, 4.$$

The depressed condition gives $s_0 = 0$. Define cyclic invariants

$$S_1 = s_1^5, \quad S_2 = s_2 s_1^3, \quad S_3 = s_3 s_1^2, \quad S_4 = s_4 s_1.$$

Once $P_1^5 = S_1$ is chosen with $P_1 \neq 0$, the remaining Fourier components are $P_2 = S_2/P_1^3$, $P_3 = S_3/P_1^2$, and $P_4 = S_4/P_1$.

The rational metacyclic basis used to express these quantities is

$$\begin{aligned} i_4 &= \sum_C x_0^2 (x_1 x_4 + x_2 x_3), & i_5 &= \sum_C x_0^3 (x_1 x_4 + x_2 x_3), \\ i_6 &= \sum_C x_0^4 (x_1 x_4 + x_2 x_3), & i_7 &= \sum_C x_0^3 (x_1^2 x_4^2 + x_2^2 x_3^2), \\ i_8 &= \sum_C x_0^4 (x_1^2 x_4^2 + x_2^2 x_3^2). \end{aligned}$$

Lean stores their values in **Invariants**. Their origin in a root tuple will be established only in a later correctness bridge; this formula module uses the exact resolvent and linear relations as its semantic interface.

6 The sextic resolvent

The discriminant of $X^5 + pX^3 + qX^2 + rX + s$ is

$$\begin{aligned} \Delta &= 108p^5s^2 - 72p^4qrs + 16p^4r^3 + 16p^3q^3s - 4p^3q^2r^2 - 900p^3rs^2 \\ &\quad + 825p^2q^2s^2 + 560p^2qr^2s - 128p^2r^4 - 630pq^3rs + 144pq^2r^3 \\ &\quad - 3750pqs^3 + 2000pr^2s^2 + 108q^5s - 27q^4r^2 + 2250q^2rs^2 \end{aligned}$$

$$-1600qr^3s + 256r^5 + 3125s^4.$$

This is `discriminant`; it was independently compared with a computer algebra discriminant on all 625 tuples $p, q, r, s \in \{-2, -1, 0, 1, 2\}$ during the source audit.

For a variable $z = i_4$, put

$$\begin{aligned} A(z) = & 2z^3 + 8rz^2 + (-6p^2r + 2pq^2 - 50qs + 24r^2)z \\ & - 15p^2qs - 16p^2r^2 + 13pq^2r + 125ps^2 - 2q^4 - 200qrs + 64r^3. \end{aligned}$$

Then Lazard's monic sextic is

$$R(z) = \left(\frac{A(z)}{2} \right)^2 - \left(z + 3r + \frac{p^2}{4} \right) \Delta. \quad (1)$$

The fraction-free equation used by the corrected Wolfram code is

$$A(z)^2 = (4z + p^2 + 12r)\Delta. \quad (2)$$

Lean's two definitions are `resolventEval` and `resolventPolynomial`. Their agreement under evaluation is the theorem `resolventPolynomial_eval`. The theorem `resolventEval_eq_zero_iff` proves the equivalence of (1) and (2).

There is a notation trap in the printed sources. Cayley's earlier invariant is

$$\Theta = (V - W)^2 = 4i_4 + p^2 + 12r,$$

whereas the section-7 sextic variable is $z = i_4$. Lean uses separate names `cayleyTheta` and `resolventCore`, preventing this collision.

For an irreducible solvable rational quintic, the sextic has a rational root. Dummit's independent F_{20} resolvent is the same invariant and gives the same criterion. A future proof must not infer specialized distinctness from generic transitivity alone: Chapman's note supplies the missing argument, and Lazard gives a separability proof for the Cayley resolvent.

7 Recovering $i_5, \dots, i_8$

After choosing i_4 , Lazard's Figure 3 gives four equations linear in $i_5, \dots, i_8$. They are transcribed as `i4SquareRhs`, `i4CubeRhs`, `i4FourthRhs`, and `i4FifthRhs`; together with the sextic equation they form `InvariantRelations`.

$$\begin{aligned} i_4^2 = & 5i_8 - 2pi_6 + 4qi_5 - 2p^2i_4 - 6p^2r + 2pq^2 + 10qs + 4r^2, \\ i_4^3 = & \frac{1}{2}[(3p^2 - 20r)i_8 + (-pq - 50s)i_7 + (-3p^3 + 28pr - 12q^2)i_6 \\ & + (3p^2q - 45ps - 6qr)i_5 + (-3p^4 + 36p^2r - 15pq^2 + 60qs - 32r^2)i_4 \\ & - 6p^4r + 3p^3q^2 + 41p^2qs + 52p^2r^2 - 54pq^2r - 250ps^2 + 14q^4 + 140qrs - 80r^3], \\ i_4^4 = & (19p^2r - 9pq^2 + 225qs - 60r^2)i_8 + (15p^2s - 8pqr + 3q^3 + 100rs)i_7 \\ & + (-4p^3r + 4p^2q^2 - 105pqs - 16pr^2 + 29q^2r + 125s^2)i_6 \\ & + (-9p^3s + 17p^2qr - 8pq^3 + 140prs + 155q^2s - 68qr^2)i_5 \\ & + (-4p^4r + 4p^3q^2 - 79p^2qs - 16p^2r^2 + 15pq^2r - 25ps^2 + 4q^4 + 80qrs)i_4 \\ & + 6p^4qs - 22p^4r^2 + 16p^3q^2r - 4p^2q^4 - 404p^2qrs + 68p^2r^3 \\ & + 132pq^3s + 42pq^2r^2 + 550prs^2 - 30q^4r - 50q^2s^2 + 20qr^2s + 16r^4, \end{aligned}$$

and

$$\begin{aligned}
2i_4^5 = & (15p^4r - 5p^3q^2 + 290p^2qs - 152p^2r^2 - 27pq^2r - 1375ps^2 + 22q^4 - 700qrs + 240r^3)i_8 \\
& + (18p^4s - 11p^3qr + 3p^2q^3 - 530p^2rs + 110pq^2s + 124pqr^2 - 41q^3r - 2375qs^2 + 200r^2s)i_7 \\
& + (-15p^5r + 5p^4q^2 - 212p^3qs + 168p^3r^2 - 83p^2q^2r + 325p^2s^2 + 10pq^4 + 1560pqrs \\
& \quad - 176pr^3 - 620q^3s - 12q^2r^2 - 1500rs^2)i_6 \\
& + (15p^4qr - 5p^3q^3 - 147p^3rs + 351p^2q^2s - 90p^2qr^2 - 43pq^3r - 3175pqs^2 \\
& \quad - 420pr^2s + 20q^5 + 215q^2rs + 152qr^3 + 625s^3)i_5 \\
& + (-15p^6r + 5p^5q^2 - 200p^4qs + 200p^4r^2 - 110p^3q^2r + 355p^3s^2 + 15p^2q^4 \\
& \quad + 1728p^2qrs - 432p^2r^3 - 752pq^3s + 220pq^2r^2 - 200prs^2 - 43q^4r + 1825q^2s^2 \\
& \quad - 2640qr^2s + 512r^4)i_4 \\
& - 30p^6r^2 + 25p^5q^2r + 198p^5s^2 - 5p^4q^4 - 491p^4qrs + 364p^4r^3 + 181p^3q^3s \\
& - 286p^3q^2r^2 - 810p^3rs^2 + 95p^2q^4r + 3005p^2q^2s^2 + 4120p^2qr^2s - 1088p^2r^4 \\
& - 12pq^6 - 4095pq^3rs + 612pq^2r^3 - 15875pqs^3 + 900pr^2s^2 + 858q^5s - 34q^4r^2 \\
& + 10700q^2rs^2 - 6240qr^3s + 960r^5 \\
& + 6250s^4.
\end{aligned}$$

Retaining these as equations, instead of embedding the enormous pre-solved rational functions, is intentional. It is exactly the finite linear-system interface needed for a later proof of existence and uniqueness.

8 The two quadratic stages

The explicit metacyclic invariants on Lazard p. 221 are:

$$\begin{aligned}
D = & 40pi_8 - 120qi_7 + (-24p^2 + 100r)i_6 + (88pq - 300s)i_5 \\
& + (-24p^3 + 100pr + 24q^2)i_4 - 80p^3r + 40p^2q^2 - 480pqs + 160pr^2 + 332q^2r + 125s^2, \\
E = & (3p^2 + 20r)i_6 + (-pq - 50s)i_5 + (3p^3 + 12pr + 3q^2)i_4 \\
& + 4p^3r - 3p^2q^2 + 40pqs + 16pr^2 - 21q^2r + 125s^2, \\
F = & (-65p^2q + 875ps - 550qr)i_8 + (-58p^2r + 41pq^2 - 275qs + 440r^2)i_7 \\
& + (85p^3q - 520p^2s - 298pqr + 366q^3 + 2100rs)i_6 \\
& + (4p^3r - 73p^2q^2 + 2095pqs - 56pr^2 - 748q^2r - 4875s^2)i_5 \\
& + (85p^4q - 418p^3s - 440p^2qr + 419pq^3 + 1590prs - 1040q^2s + 524qr^2)i_4 \\
& - 12p^5s + 158p^4qr - 85p^3q^3 - 1462p^3rs - 159p^2q^2s + 142p^2qr^2 \\
& + 896pq^3r + 175pqs^2 + 2900pr^2s - 402q^5 - 1925q^2rs - 448qr^3 - 1875s^3, \\
G = & (-35p^2q - 250ps - 200qr)i_8 + (-22p^2r + 19pq^2 + 650qs - 40r^2)i_7 \\
& + (15p^3q + 195p^2s + 68pqr - 6q^3 - 1100rs)i_6 \\
& + (-4p^3r - 27p^2q^2 - 270pqs + 96pr^2 - 182q^2r + 3000s^2)i_5 \\
& + (15p^4q + 213p^3s + 50p^2qr + pq^3 - 940prs + 515q^2s - 184qr^2)i_4 \\
& + 12p^5s + 42p^4qr - 15p^3q^3 + 492p^3rs - 156p^2q^2s + 358p^2qr^2 \\
& - 246pq^3r + 2825pqs^2 - 1400pr^2s + 42q^5 + 550q^2rs - 232qr^3 - 1250s^3.
\end{aligned}$$

They are `invariantD`, `invariantE`, `invariantF`, and `invariantG`.

Choose ε_0 with $\varepsilon_0^2 = 5D$. The corrected choice is

$$\varepsilon = \begin{cases} -\varepsilon_0, & E + F/\varepsilon_0 = 0, \\ \varepsilon_0, & E + F/\varepsilon_0 \neq 0. \end{cases} \quad (3)$$

This is literally `correctEpsilon`. Lean proves `correctEpsilon_sq`. Under the sourced assertion that at least one of $E + F/\varepsilon_0$ and $E - F/\varepsilon_0$ is nonzero, the theorem

`correctedTRadicand_ne_zero`

proves that the corrected radicand is nonzero. One then chooses

$$T^2 = \frac{5}{2} \left(E + \frac{F}{\varepsilon} \right), \quad U = \frac{5G}{T\varepsilon}. \quad (4)$$

The factor $5/2$ is the one used in section 7, Maple, and both Wolfram ports; an earlier prose passage in Lazard omits the relevant scaling.

The records `EpsilonChoice`, `TChoice`, and `RadicalCertificate` expose the equations and nondegeneracy assumptions. In particular, the certificate stores the complementary equation $U^2 = \frac{5}{2}(E - F/\varepsilon)$. The theorem `RadicalCertificate.t_nonzero` derives $T \neq 0$ from the corrected radicand rather than trusting total field division, and `RadicalCertificate.initial_relations` proves $TU\varepsilon = 5G$ from the definition of U .

9 The fifth-root radicand and coherent signs

Define

$$\begin{aligned} H &= 25(2i_5 - pq - 5s), \\ I &= 25[40pi_8 - 70qi_7 + (-24p^2 + 100r)i_6 + (68pq - 300s)i_5 \\ &\quad + (-24p^3 + 100pr - 46q^2)i_4 - 80p^3r + 20p^2q^2 - 255pqs + 160pr^2 - 28q^2r + 125s^2], \\ J &= -25pi_8 - 25qi_7 + (-9p^2 - 60r)i_6 + (-7pq + 525s)i_5 \\ &\quad + (-p^3 - 96pr + 11q^2)i_4 + 50p^3r - 7p^2q^2 - 145pqs - 308pr^2 + 128q^2r - 1000s^2, \\ K &= -125pi_8 + 75qi_7 + (67p^2 - 420r)i_6 + (-109pq + 1175s)i_5 \\ &\quad + (63p^3 - 412pr + 27q^2)i_4 + 210p^3r - 79p^2q^2 - 415pqs - 676pr^2 + 496q^2r - 750s^2. \end{aligned}$$

These are `invariantH` through `invariantK`. For a coherent triple (ε, T, U) , set

$$Q_1 = \frac{5}{4} \left(H + \frac{I}{\varepsilon} + \frac{TJ + UK}{E} \right). \quad (5)$$

The four allowed triples are

$$(\varepsilon, T, U), \quad (\varepsilon, -T, -U), \quad (-\varepsilon, U, -T), \quad (-\varepsilon, -U, T).$$

They are the four constructors handled by `branchTriple`. The selector `selectSignBranch` returns the first branch with $Q_1 \neq 0$, or `none` if the mathematical existence hypothesis has not been supplied. Its theorem `selectSignBranch_sound` proves that a returned branch is indeed nonzero. Quadratic coherence is the separate theorem

`QuadraticRelations.of_branchTriple`.

It proves that each transformation preserves both square equations and $TU\varepsilon = 5G$. A `RadicalCertificate` stores the chosen branch, the nonzero fact, and one shared P_1 satisfying $P_1^5 = Q_1$; Lean derives $P_1 \neq 0$ in `RadicalCertificate.p1_nonzero`.

This finite branch mechanism is the algebraic content of the corrected algorithm. Domen's later use of Mathematica's `Together` merely repairs syntactic evaluation of fractional powers during root matching. Since Lean uses a single ω and field expressions, it has no mathematical analogue here.

10 The corrected P_{22} and Fourier reconstruction

Lazard's remaining numerators are transcribed one per Lean definition:

$$\begin{aligned}
P_{41} &= -5p, \\
P_{42} &= 5(10i_7 - 4pi_5 - 14qi_4 - 4p^2q + 45ps - 72qr), \\
P_{31} &= -25q, \\
P_{32} &= 25(-10i_8 + 2pi_6 - 22qi_5 + 2p^2i_4 + 20p^2r + 2pq^2 - 35qs - 40r^2), \\
P_{33} &= 5(35i_8 - 4pi_6 + 23qi_5 + (-6p^2 + 12r)i_4 - 58p^2r + 14pq^2 - 105qs + 76r^2), \\
P_{34} &= 5(5i_8 - 22pi_6 + 14qi_5 + (-18p^2 + 16r)i_4 - 34p^2r + 22pq^2 - 140qs + 68r^2), \\
P_{21} &= 5(3i_4 + 2p^2 - 16r), \\
P_{22} &= 25(-10qi_6 + (8p^2 - 50r)i_5 + (-2pq - 25s)i_4 \\
&\quad + \boxed{8p^3q} + \boxed{70q^3} - 20p^2s - 26pqr + 50rs), \\
P_{23} &= 25(-4pi_7 - qi_6 + 4ri_5 + (-3pq + 15s)i_4 + 26p^2s - 26pqr + 7q^3 - 40rs), \\
P_{24} &= 25(3pi_7 - 18qi_6 + 22ri_5 + (-14pq + 20s)i_4 + 18p^2s - 33pqr + 21q^3 + 30rs).
\end{aligned}$$

The boxed terms are Reshetnikov's corrections. Lazard p. 222 prints $8p^3$ and $70q^3q$; the corrected terms are independently confirmed by the original Maple worksheet, which has $200p^3q$ and $1750q^3$ after distributing the outer factor 25. Lean makes the corrected formula conspicuous in `p22`.

With the selected (ε, T, U) and shared P_1 , define

$$\begin{aligned}
P_4 &= \frac{P_{41}}{2P_1} + \frac{P_{42}}{2\varepsilon P_1}, \\
P_3 &= \frac{P_{31}}{4P_1^2} + \frac{P_{32}}{4\varepsilon P_1^2} + \frac{P_{33}T + P_{34}U}{10EP_1^2}, \\
P_2 &= \frac{P_{21}}{4P_1^3} + \frac{P_{22}}{4\varepsilon P_1^3} + \frac{P_{23}T + P_{24}U}{10EP_1^3}.
\end{aligned}$$

These are `fourierP4`, `fourierP3`, and `fourierP2`.

For one primitive fifth root ω , all five depressed candidates are

$$y_k = \frac{\omega^k P_1 + \omega^{2k} P_2 + \omega^{3k} P_3 + \omega^{4k} P_4}{5}, \quad k = 0, 1, 2, 3, 4. \quad (6)$$

The original roots are

$$x_k = y_k - \frac{b}{5a}.$$

Lean represents ω by `FifthRootOfUnity`; its field `primitive` certifies that `value` is a primitive fifth root. The single indexed formula `solveDepressed` replaces five branch-sensitive Mathematica expressions. `solveGeneral` performs the final translation, and `solveRational` keeps the rational invariant certificate visible.

11 Lean-to-paper crosswalk

Lean declaration	Lazard notation/source	Role
<code>GeneralQuintic</code> , <code>GeneralQuintic.polynomial</code> <code>GeneralQuintic.polynomial_eval</code>	a, b, c, d, e, f evaluation at x	Original polynomial and its coefficient representation. Proved agreement of polynomial and scalar evaluation.
<code>DepressedQuintic</code> , <code>depress</code> <code>depress_eval</code> <code>discriminant</code>	p, q, r, s ; p. 219 $X = Y - b/(5a)$ Δ ; pp. 219–220	Normalize and depress. Kernel-checked translation identity. Explicit depressed-quintic discriminant.
<code>resolventCore</code>	$A(z)$	Cubic whose square enters the sextic.
<code>cayleyTheta</code>	$\Theta = 4i_4 + p^2 + 12r$; p. 219	Keeps Cayley Θ distinct from $z = i_4$.
<code>resolventPolynomial</code> <code>resolventEval_eq_zero_iff</code>	$R(z)$; pp. 219–220 fraction-free Stack Exchange equation	Monic degree-six polynomial. Proved equivalence of the two displays.
<code>Invariants</code> <code>i4SquareRhs–i4FifthRhs</code> <code>InvariantRelations</code> , <code>InvariantRelations.map</code>	$i_4, \dots, i_8$; pp. 217–220 Figure 3; pp. 220–221 sextic plus Figure 3	Metacyclic invariant values. Linear system for $i_5, \dots, i_8$. Exact rational-stage certificate, proved stable under scalar extension.
<code>invariantD–invariantG</code>	D, E, F, G ; p. 221	Quadratic-stage invariant polynomials.
<code>correctEpsilon</code>	$\varepsilon \mapsto -\varepsilon$ if $T = 0$; p. 221 and Zapodovnikov	Corrected first sign choice.
<code>squareBranchAvailable</code>	not both $E \pm F/\varepsilon_0$ zero	Explicit nondegeneracy obligation.
<code>correctedTRadicand_ne_zero</code>	p. 221 nonvanishing argument	Proved consequence of the obligation.
<code>radicalU</code> <code>invariantH–invariantK</code>	$U = 5G/(T\varepsilon)$ H, I, J, K ; pp. 221–222	Second quadratic-stage value. Projections defining Q_1 .
<code>QuadraticRelations</code>	equations for ε, T, U	Records both squares and $TU\varepsilon = 5G$.
<code>SignBranch</code> , <code>branchTriple</code> <code>QuadraticRelations.of_</code> <code>branchTriple</code> <code>selectSignBranch</code>	four coherent triples coherent sign action paper’s $Q_1 \neq 0$ choice	Finite shared-sign orbit. Proves all four transformations preserve the equations. Honest <code>Option</code> ; no unsafe list indexing.
<code>q1</code> <code>p22</code>	Q_1 corrected P_{22} ; p. 222/MSE	Fifth-root radicand. Contains $8p^3q$ and $70q^3$.
<code>fourierP2–fourierP4</code> <code>RadicalCertificate</code>	P_2, P_3, P_4 ; p. 222 shared ε, T, U, P_1	Remaining Fourier components. Equations and nonzero obligations stored once.
<code>RadicalCertificate.initial_</code> <code>relations</code> , <code>RadicalCertificate.</code> <code>chosen_relations</code>	quadratic coherence	Connect the certificate to both the initial and selected triples.
<code>FifthRootOfUnity</code> <code>solveDepressed</code>	primitive ω inverse Fourier formula	Shared fifth root of unity. All five depressed candidates.
<code>solveGeneral</code> <code>RationalInvariantCertificate</code>	$x_k = y_k - b/(5a)$ rational $i_4, \dots, i_8$	Undo translation. A supplied witness over $\mathbb{Q}$; it is not a search procedure.

Lean declaration	Lazard notation/source	Role
<code>FormulaSound</code> , <code>RationalFormulaSound</code>	pointwise root identity	Unproved propositions; not axioms and not asserted.
<code>FormulaFactors</code> , <code>RationalFormulaFactors</code>	full five-factor identity	Stronger unproved completeness targets.

12 Why the certificates are part of the algorithm

The following denominators occur in the formula:

$$a, \quad \varepsilon, \quad T, \quad E, \quad P_1.$$

The formalization treats their nonvanishing as follows.

- $a \neq 0$ is an explicit hypothesis of `depress_eval` and of the future soundness and factorization targets.
- The field `epsilon0_nonzero` records the sourced irreducible-case fact; the theorem `epsilon0_nonzero` in the `RadicalCertificate` namespace proves that sign correction preserves it.
- `epsilon_branches`, the square equation for T , and `correctedTRadicand_ne_zero` imply $T \neq 0$.
- The stored square equation for U , together with the definition $U = 5G/(T\varepsilon)$, yields `RadicalCertificate.initial_relations`; `QuadraticRelations.of_branchTriple` proves that all four coherent sign choices preserve the two square equations and $TU\varepsilon = 5G$.
- `invariantE_nonzero` records Lazard’s $E = T^2 + U^2 \neq 0$ argument in the rational case.
- `branch_nonzero` and $P_1^5 = Q_1$ imply $P_1 \neq 0$.

This organization is intentionally relational. A later executable algebraic- number layer may compute the certificates, but the mathematical formula is not made dependent on a particular computer algebra system’s analytic square-root or fractional-power convention.

13 What remains to prove

The generic target `FormulaSound` says that, for a valid leading coefficient, an invariant tuple satisfying `InvariantRelations`, a valid radical certificate, and a primitive fifth root of unity, every indexed value from `solveGeneral` annihilates the original quintic. Its rational specialization `RationalFormulaSound` additionally requires that the invariant tuple be supplied and certified in $\mathbb{Q}$ before scalar extension. Neither proposition asserts that such witnesses can be found.

Pointwise soundness alone does not express that all five roots, with their multiplicities, have been recovered. The separate targets `FormulaFactors` and `RationalFormulaFactors` state the stronger polynomial identity

$$\text{GeneralQuintic.polynomial} = a \prod_{k=0}^4 (X - \text{solveGeneral}(k)).$$

Thus the API cleanly separates conditional root soundness, five-root completeness, and the still-absent existence/search layer.

A complete proof can be decomposed into the following independent lemmas:

1. identify the sextic with the F_{20} orbit resolvent and prove the rational-root solvability criterion;
2. prove specialized separability/distinctness, using Lazard’s Cayley argument or Chapman’s repair rather than the invalid transitivity shortcut;
3. derive the Figure 3 linear system from the root-defined invariants and prove its determinant nonzero;
4. derive the displayed $D, \dots, K$ reductions;
5. prove that the two quadratic stages recover the cyclic projections;
6. prove that one of the four coherent Q_1 values is nonzero;
7. identify $P_1, \dots, P_4$ with the four Fourier sums; and
8. apply Fourier inversion and `depress_eval`.

The factorization target then gives exhaustiveness and, under irreducibility and separability, pairwise distinctness. None of these unfinished statements is admitted with `axiom`, `sorry`, or an opaque external oracle.

14 Regression evidence and source cautions

The wider Smithereens review found useful future regression cases, but also showed why notebook output is not a proof oracle.

- The corrected zero- T branch is exercised by $X^5 - 10X^2 - 10X - 6$, for which the audited source gives $i_4 = 30$.
- An older direct notebook records a failure on $X^5 - 5X - 12$ when it omits an adequate branch fallback. This is a valuable all-five-roots case.
- Reshetnikov’s strongest stored exact harness checks five values, substitution residuals, and distinctness for six polynomials, including the very large target polynomial from the Stack Exchange question.
- Some other notebook runs abort after ten cases, some use `PossibleZeroQ`, and the independent package checks high-precision residuals without proving symbolic identities or complete root coverage.
- The label of $X^5 + 20X + 32$ as F_{20} in one newer document is incorrect; its group is dihedral. It should not be used as an F_{20} classification fixture.
- Dummit’s printed examples have later corrections, and his original factorization proof needs Chapman’s specialized-distinctness lemma.

Consequently the committed Lean module uses source formulas and algebraic certificates, not notebook success labels. Future regression computations are useful checks against transcription errors, but kernel-checked symbolic identities will remain the correctness criterion.

15 Reproducibility and audit

From the repository root, the module and its public import surface are checked with

```
lake env lean \
  Algebra/PolynomialFormulas/Lean/PolynomialFormulas/LazardQuintic.lean
```

```
LEAN_NUM_THREADS=0 lake build +PolynomialFormulas
```

The document is rendered by first changing to its directory and then running the following command three times (for the table of contents, citations, and cross-references):

```
cd Algebra/PolynomialFormulas
pdflatex -interaction=nonstopmode -halt-on-error \
  LazardQuinticFormalization.tex
```

For the present contribution, an audit should additionally verify:

1. no occurrence of `sorry`, `admit`, or a new `axiom`;
2. the displayed corrected terms occur as $8*c.p^3*c.q$ and $70*c.q^3$;
3. `correctEpsilon` precedes construction of T and U ;
4. `solveDepressed` uses one stored P_1 and one primitive ω ; and
5. the PDF is built from the committed L^AT_EX source without errors.

References

- [1] Daniel Lazard, *Solving Quintics by Radicals*, in Olav Arnfinn Laudal and Ragni Piene (eds.), *The Legacy of Niels Henrik Abel*, Springer, 2004, pp. 207–225. https://doi.org/10.1007/978-3-642-18908-1_6.
- [2] David S. Dummit, *Solving Solvable Quintics*, *Mathematics of Computation* 57 (1991), no. 195, 387–401.
- [3] Robin Chapman, *Note on Solving Solvable Quintics*, one-page correction note hosted with Dummit’s source materials. https://site.uvm.edu/ddummit/files/2021/04/Solving_Solvable_Quintics_note_by_Robin_Chapman.pdf.
- [4] Vladimir Reshetnikov and Valeriy Zapodovnikov, *Solving quintic in radicals*, *Mathematica Stack Exchange* question 133550 and answers, 2016–2025. <https://mathematica.stackexchange.com/questions/133550/solving-quintic-in-radicals>.